

03 — Step-by-Step Implementation



DOCUMENT PURPOSE

The complete, reproducible build record for the deployed system: every phase, every command, the justification for non-default choices, a 30-point verification checklist, and a disaster-recovery runbook. This document is the evidence base for the capability claims made in documents 01, 02 and 05.

Previous: [02-Architecture-and-Hardware](#) · **Next:** [04-Interactive-Demo-Guide](#)

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1. Preconditions

1.1 Required items

#	Item	Note
1	Raspberry Pi 5 (8 GB), assembled with cooler, NVMe, and 27 W supply	Per BOM

#	Item	Note
2	Provisioning workstation with Raspberry Pi Imager	https://www.raspberrypi.com/software/
3	microSD card, 64 GB, A2 class	OS only
4	Cat6 cable	Preferred for all build and demonstration operations
5	Keyboard and monitor (optional)	Recovery access if SSH is unavailable
6	USB-C data cable (optional)	Direct device-to-device transfer
7	USB removable drive	Off-site backup target

1.2 Build invariants

⚡ THE FIVE CONSTRAINTS THAT GOVERN THE ENTIRE PROCEDURE

1. **The microSD carries the operating system only.** All user data is written to the NVMe. Every deviation from this invalidates the recoverability claim (P1).
2. **Docker's root directory remains on the microSD.** Enforced and verified in §6.3.
3. **Encryption is enabled before any user data is imported** (§9). Encryption applies to newly written files; pre-existing files are not retroactively encrypted.
4. **A 5 V / 5 A supply is used.** Under-voltage is a documented cause of non-deterministic SD write corruption, which typically manifests days after the build.
5. **Every image tag is pinned.** `:latest` permits an unreviewed version change during any subsequent pull. Pinning is also what makes rollback possible (§14, DR-5).

1.3 Notation

```
# A command to execute
# → output or expected result
```

🔗 VERSION VERIFICATION

The commands recorded here were executed on **2026-09-30**. Install strings and image tags for CasaOS, Nextcloud, and Jellyfin are published in their respective official documentation. Version-sensitive values are marked inline. Verification of the current upstream command before repeating this build is recommended.

2. Phase 1 — Base OS provisioning

Objective: produce a booted, network-reachable host with a correct hostname and no application content.

2.1 Image selection

Raspberry Pi Imager → device *Raspberry Pi 5* → OS *Other* → **Raspberry Pi OS (64-bit)**, Debian Bookworm.

EDITION SELECTION

The standard 64-bit image is used. The Desktop environment is not required — the host is headless by design and the desktop environment consumes approximately 400 MB of additional RAM. Lite would be sufficient but omits several utilities referenced below.

2.2 Provisioning options

 **Advanced options** (Ctrl+Shift+X) — all fields are set deliberately:

Tab	Field	Value	Basis
General	Hostname	pi-cloud	Becomes pi-cloud.local and the default service hostname
General	Username / password	Dedicated non-default account	A named unprivileged account; direct root access is disabled in §4.2
General	Wireless SSID / password	Deployment network	Wireless operation only
General	Wireless country	Deployment country	Unset country codes constrain radio power to the lowest common regulatory limit and may constitute non-compliant operation
Services	Enable SSH	Enabled	Administrative access
Options	Set hostname	Enabled	
Options	Enable wait-for-network	Enabled	Prevents a headless host from booting before DHCP completes — a common cause of unreachable first boots

2.3 Write, boot, verify

Write with verification enabled. Insert the card, connect Ethernet, attach the NVMe, then apply power. Allow approximately 40 seconds; a steady green activity LED indicates a completed boot.

```
# Address discovery, in order of reliability
nslookup pi-cloud.local

arp -a

# or consult the router's DHCP client table
```

⚠ ETHERNET IS SPECIFIED FOR THE BUILD

A wired host completes network initialisation in a few seconds. Over a congested or enterprise wireless network, a headless host may fail DHCP entirely and self-assign a link-local address, presenting as an unreachable device. Wireless client isolation, where enabled on institutional networks, produces the same symptom.

3. Phase 2 — Host initialisation

```
ssh admin@pi-cloud.local
```

3.1 Hardware enumeration

```
lscpu | grep -E "Model name|Architecture"

free -h

lsblk

nvme list

lspci | grep -i non-volatile
```

Expected: 4 × Cortex-A76, `aarch64`; ≈7.6 GiB total memory; the NVMe device present in `lsblk`; one `Non-Volatile memory controller` line from `lspci`.

⚡ THE SINGLE MOST COMMON BUILD FAILURE

An empty result from `lspci | grep -i non-volatile` means the PCIe device has not enumerated. The cause is almost always a partially seated FFC cable rather than a defective SSD. Power down, reseal the cable observing the minimum bend radius, and re-verify. Diagnosing this in hardware before proceeding avoids a multi-hour software debugging exercise against a device that is not present.

3.2 System and firmware update

```
sudo apt update && sudo apt full-upgrade -y  
sudo rpi-eeprom-update -a  
sudo reboot
```

```
ssh admin@pi-cloud.local
```

RATIONALE FOR THE EEPROM UPDATE

The Pi 5's bootloader resides in on-board EEPROM, independent of the microSD. An outdated bootloader with a current OS produces intermittent PCIe, USB, and storage enumeration faults that are indistinguishable from hardware failure. This step is 30 seconds and eliminates that class of fault.

3.3 System identity

```
sudo timedatectl set-timezone Asia/Kolkata  
timedatectl status  
hostnamectl  
sudo raspi-config --expand-rootfs && sudo reboot
```

```
# Static address: configure a DHCP reservation at the router  
# MAC address of the host → 192.168.1.50
```

RESERVATION OVER LOCAL STATIC CONFIGURATION

A DHCP reservation maintained at the router is a single point of configuration and cannot conflict with another device. Local static configuration in `dhcpcd.conf` or NetworkManager creates two sources of truth for an address intended to remain fixed over a multi-year deployment.

3.4 Service discovery

```
sudo systemctl enable --now avahi-daemon  
ping -c2 pi-cloud.local
```

```
sudo nano /etc/hosts
```

```
# 192.168.1.50    pi-cloud    cloud.home
```

4. Phase 3 — Host hardening

Objective: a hardened host before any application is installed or reachable.

4.1 Administrative account

```
sudo adduser piuser
sudo usermod -aG sudo piuser
sudo visudo          # confirm membership
```

4.2 SSH

```
# Workstation:
ssh-copy-id piuser@pi-cloud.local
# Verify key authentication before continuing
ssh piuser@pi-cloud.local
```

```
# Host:
sudo nano /etc/ssh/sshd_config
```

```
PermitRootLogin no
PasswordAuthentication no
KbdInteractiveAuthentication no
PubkeyAuthentication yes
X11Forwarding no
MaxAuthTries 3
LoginGraceTime 30
```

```
sudo systemctl restart ssh
```

MANDATORY VERIFICATION BEFORE SESSION CLOSURE

Disabling password authentication before confirming that key authentication functions correctly removes the only administrative access path and requires physical intervention to recover. **Key authentication must be confirmed in a second, independent session before the active session is closed.**

4.3 Firewall

```
sudo apt install -y ufw
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp    comment 'SSH'
sudo ufw allow 80/tcp    comment 'CasaOS'
sudo ufw allow 81/tcp    comment 'CasaOS HTTPS'
sudo ufw allow 8080/tcp  comment 'Nextcloud'
sudo ufw allow 8096/tcp  comment 'Jellyfin'
sudo ufw allow 445/tcp   comment 'Samba'
sudo ufw allow 139/tcp   comment 'NetBIOS'
sudo ufw allow 2049/tcp  comment 'NFS'
sudo ufw enable
sudo ufw status verbose
```

APPLICABILITY

A packet filter is one of several required layers for an internet-facing service. In this architecture it functions as a second, auditable layer behind the router's NAT. The allowlist is the complete, reviewed set — a port is added only where a deployed service requires it.

4.4 Automated security patching

```
sudo apt install -y unattended-upgrades
sudo dpkg-reconfigure -pnow unattended-upgrades
sudo nano /etc/apt/apt.conf.d/50unattended-upgrades
```

```
Unattended-Upgrade::Allowed-Origins {
    "${distro_id}:${distro_codename}-security";
```

```
};
```

SCOPE OF AUTOMATION

Security-channel patches are applied automatically. Distribution version upgrades and application image updates are excluded and are performed deliberately (DR-5). An unmonitored major version change on an unattended host is a material availability risk.

4.5 Thermal and power state

```
vcgencmd measure_temp      # expected: well below 80
vcgencmd get_throttled      # expected: throttled=0x0
```

INTERPRETING THE THROTTLE MASK

The output is a bitmask; `0x0` indicates a healthy state. A non-zero value containing `throttle_undervoltage` indicates a power supply fault, not a software condition, and produces intermittent faults attributed to storage or USB. Where performance is inconsistent and unexplained, this is the first check.

5. Phase 4 — NVMe storage

Objective: provision the persistent data volume, enforce invariant 1 (§1.2), and confirm the mount survives reboot and disk absence.

5.1 Confirm PCIe enumeration

```
lspci -nn | grep -i non-volatile
```

5.2 Partitioning

```
lsblk -o NAME,SIZE,TYPE,MODEL
sudo fdisk /dev/nvme0n1
```

`fdisk` session:

```
g          # GPT partition table
```



```
n          # new partition
1          # partition number
↵ ↵      # first sector (1 MiB alignment)
↵ ↵      # last sector (use all remaining space)
w          # write
q          # quit
```

5.3 Filesystem creation

```
sudo wipefs -a /dev/nvme0n1p1      # remove any pre-existing signatures
sudo mkfs.ext4 -L picloud /dev/nvme0n1p1
```

PARTITION-LEVEL OPERATIONS

All subsequent commands reference the partition, not the whole device. This prevents accidental formatting of the wrong target and preserves the possibility of adding a partition without reformatting.

5.4 Persistent mounting

```
sudo mkdir -p /mnt/storage
sudo blkid /dev/nvme0n1p1          # record UUID
```

```
sudo nano /etc/fstab
```

```
UUID=<uuid> /mnt/storage ext4 defaults,noatime,nofail,discard 0 0
```

```
sudo mount -a
findmnt /mnt/storage
df -h /mnt/storage
```

NOFAIL IS MANDATORY ON A DATA VOLUME

Without `nofail`, an absent or failed data disk causes the host to block in an emergency boot state, which is unreachable over the network. This converts a replaceable commodity component into an on-site intervention. The host must always boot, so that it can be reached and diagnosed.

5.5 Ownership and permissions

```
sudo chown -R 1000:1000 /mnt/storage
sudo chmod 755 /mnt/storage
ls -ln /mnt/storage          # numeric IDs – the authoritative value
```

⚠ NUMERIC UID MISMATCH – HIGHEST-FREQUENCY APPLICATION DEFECT

A container running as a different unprivileged UID cannot write to a directory it does not own, and the resulting application error is typically a generic authorisation failure that directs diagnosis toward the application rather than the host.

```
ls -ln /mnt/storage/nextcloud-data          # read the numbers, not the names
sudo chown -R 33:33 /mnt/storage/nextcloud-data  # match the container's user
```

A single UID/GID policy is applied across all services and declared explicitly in each Compose file. Ambiguity at this layer is the most common cause of reported application faults.

5.6 Directory structure

```
sudo mkdir -p /mnt/storage/{nextcloud-db,nextcloud-data,jellyfin-config,jellyfin-
cache,shares,snapshots,backups}
sudo mkdir -p /mnt/storage/media/{movies,tv,music,photos}
sudo chown -R 1000:1000 /mnt/storage
```

5.7 Health instrumentation

```
sudo tee /usr/local/bin/picloud-health <<'EOF'
#!/usr/bin/env bash
# PI-CLOUD health check – non-destructive, safe to run at any time
set -u
echo "=== PI-CLOUD HEALTH ==="
echo "--- uptime / load ---"; uptime
echo "--- thermals / throttling ---"
vcgencmd measure_temp; echo -n "throttle mask: "; vcgencmd get_throttled
echo "--- memory ---"; free -h
```

```

echo "--- filesystems ---"; df -h / /mnt/storage
echo "--- data volume ---"; findmnt /mnt/storage || echo "!! /mnt/storage NOT MOUNTED"
echo "--- containers ---"
docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}' 2>/dev/null || echo
"docker unavailable"
echo "--- listening ports ---"; sudo ss -tlnp
echo "--- storage health ---"
sudo smartctl -H /dev/nvme0n1 2>/dev/null || echo "smartctl: no report"
EOF
sudo chmod +x /usr/local/bin/picloud-health
picloud-health

```

PURPOSE

A single non-destructive command that reports load, thermal and power state, memory, filesystem utilisation, mount state, container state, listening services, and storage health. It is the first diagnostic executed for any anomaly, and the primary evidence artefact for the verification record.

5.8 Optional — boot from NVMe

```

sudo apt install -y raspberrypi-nvme
sudo raspi-config      # Advanced Options → Boot Order → NVMe first

```

NOT ADOPTED

Keeping the OS on a removable card makes invariant 1 (§1.2) explicit and self-evident, and means a filesystem fault cannot render the device unbootable. The trade-off is accepted in favour of the recovery property.

6. Phase 5 — Container runtime

6.1 Installation from the upstream repository

```

sudo apt install -y ca-certificates curl gnupg
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/debian/gpg \
| sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

```

```

sudo chmod a+r /etc/apt/keyrings/docker.gpg

echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/debian bookworm stable" \
| sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt update

sudo apt install -y docker-ce docker-ce-cli containerd.io \
                docker-buildx-plugin docker-compose-plugin

sudo docker run hello-world

docker --version && docker compose version

```

```

sudo usermod -aG docker piuser && newgrp docker && docker ps

```



DOCKER GROUP MEMBERSHIP IS EQUIVALENT TO ROOT

Access to the Docker socket permits mounting the host filesystem into a container and therefore confers host root. This is an accepted trade for administrative convenience on a single-operator appliance. On a shared or multi-tenant host, `sudo docker` is required instead. Recorded here as a declared property of the configuration, not an oversight.

6.2 Daemon configuration

```

sudo mkdir -p /etc/docker

sudo nano /etc/docker/daemon.json

```

```

{
  "log-driver": "json-file",
  "log-opts": { "max-size": "10m", "max-file": "3" },
  "live-restore": true,
  "default-address-pools": [
    { "base": "172.20.0.0/16", "size": "24" }
  ]
}

```

Key	Rationale
<code>log-opts</code>	Prevents unbounded log growth from exhausting the microSD. Container log exhaustion is a documented and frequently encountered self-hosted service outage
<code>live-restore</code>	Containers continue running across daemon restarts and unattended upgrades, removing a recurring short-duration availability event
<code>default-address-pools</code>	Prevents the Docker subnet from colliding with the LAN or a VPN allocation, a fault that is difficult to diagnose after the fact

```
sudo systemctl restart docker && sudo systemctl enable docker
```

6.3 Verification of invariant 2 (§1.2)

```
docker info | grep -E "Docker Root Dir|Storage Driver"
# Docker Root Dir: /var/lib/docker      ← must be on the microSD
findmnt /var/lib/docker
```

⚡ MANDATORY GATE

If `Docker Root Dir` resolves beneath `/mnt/storage`, container image layers, logs, and transcode caches share the volume with user data. Volume exhaustion will cause write failures affecting stored data. **Execution must not proceed past this point until corrected.**

6.4 Project structure and secret material

```
sudo mkdir -p /opt/picloud/{nextcloud,jellyfin}
sudo chown -R 1000:1000 /opt/picloud
```

```
sudo nano /opt/picloud/nextcloud/.env
```

```
# Secrets — generated, not composed by hand
DB_PASSWORD=$(openssl rand -base64 32)
NC_ADMIN_PASSWORD=$(openssl rand -base64 24)
NC_ADMIN_USER=<administrator>
```

```
sudo chmod 600 /opt/picloud/nextcloud/.env
sudo chown 1000:1000 /opt/picloud/nextcloud/.env
```

SECRET HANDLING

Generated entropy rather than human-composed values. The file is mode `600`, owned by the service account, located outside any shared path, and excluded from version control. These four properties constitute the control; a password alone does not.

7. Phase 6 — CasaOS

7.1 Installation

```
curl -fsSL https://get.casa.os | sudo bash
```

UPSTREAM COMMAND VERIFICATION

CasaOS has published its installer under both `get.casa.os` and `get.casaos.io`. If the recorded domain does not resolve, the current command must be taken from the official CasaOS documentation. Piping a remotely fetched script to a privileged shell is not an operation to be improvised, and the domain must be verified against the vendor's published source.

The installer detects the existing Docker installation and adopts it, installs the CasaOS services and application-management daemon, and exposes the dashboard.

7.2 Verification

```
systemctl status casaos --no-pager  
docker ps --format 'table {{.Names}}\t{{.Status}}'  
ss -tlnp | grep -E ':80|:81'
```

```
http://pi-cloud.local
```

⚡ CONFIGURATION HAZARD — DECLINED EXPLICITLY

CasaOS may offer to install the operating system onto an attached data drive.

This offer is declined. Docker's root must remain on the microSD per invariant 2 (§1.2). Accepting it places image layers, logs, and caches on the volume containing user data, and produces volume exhaustion and write failures. This is the most damaging single action available within CasaOS, and it is presented as a convenience.

7.3 Post-installation configuration

1. Create a **local** administrator account. No online account is required or used.
2. Dashboard → confirm CPU, memory, and both volumes, with `/mnt/storage` correctly identified.
3. Settings → confirm `/mnt/storage` is recognised as the data volume.
4. **Disable automatic application updates**, or restrict to notification only.

Upgrades are performed deliberately under supervision (DR-5). An unmonitored version change is a material availability risk.

5. **Do not deploy applications from the application catalogue.**

The catalogue abstracts the image tag, volume mappings, environment variables, and network mode — the four parameters that determine whether the security and recoverability requirements in document 02 are satisfied. CasaOS is retained as the management and monitoring plane; applications are declared in version-controlled Compose files.

8. Phase 7 — Nextcloud

Objective: Nextcloud with a production-appropriate database, reachable on the LAN, with encryption enabled before any data is written.

8.1 Service declaration

```
sudo nano /opt/picloud/nextcloud/docker-compose.yml
```

```
#
```

```

# PI-CLOUD - Nextcloud

#   · PostgreSQL 16 (not SQLite - see rationale table)
#   · Image tag pinned; :latest is never used
#   · All persistent state on the NVMe; Docker root remains on the microSD
# =====

name: picloud-nextcloud

services:
  db:
    image: postgres:16-alpine
    container_name: nc-db
    restart: unless-stopped
    volumes:
      - /mnt/storage/nextcloud-db:/var/lib/postgresql/data
    environment:
      POSTGRES_DB: nextcloud
      POSTGRES_USER: nextcloud
      POSTGRES_PASSWORD: ${DB_PASSWORD}
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U nextcloud -d nextcloud"]
      interval: 10s
      timeout: 5s
      retries: 10
      start_period: 30s
    networks: [nc-net]
    # No published ports - the database is addressable only from the
    # application container over the internal bridge network.

  app:
    image: nextcloud:29-apache # PINNED. Changed deliberately only.
    container_name: nextcloud
    restart: unless-stopped
    depends_on:
      db:
        condition: service_healthy
    ports:
      - "192.168.1.50:8080:80" # bound to the LAN address, not 0.0.0.0

```



```

volumes:
  - /mnt/storage/nextcloud-data:/var/www/html

environment:
  POSTGRES_HOST: db
  POSTGRES_DB: nextcloud
  POSTGRES_USER: nextcloud
  POSTGRES_PASSWORD: ${DB_PASSWORD}
  NEXTCLOUD_ADMIN_USER: ${NC_ADMIN_USER}
  NEXTCLOUD_ADMIN_PASSWORD: ${NC_ADMIN_PASSWORD}
  NEXTCLOUD_TRUSTED_DOMAINS: "192.168.1.50 pi-cloud.local cloud.home"
  NEXTCLOUD_DATA_DIR: /var/www/html/data

networks: [nc-net]

networks:
  nc-net:
    driver: bridge

```

Declaration	Rationale
<code>postgres:16-alpine</code>	SQLite under concurrent access over shared or network storage is a documented corruption source. PostgreSQL fits the memory budget (document 02 §4) and provides superior behaviour under write concurrency
<code>nextcloud:29-apache</code>	A version, not a moving reference. Pinning makes the system reproducible and rollback possible (DR-5)
<code>192.168.1.50:8080:80</code>	Service is not listening on any interface other than the LAN address
No <code>ports: on db</code>	The database is unreachable from the LAN and from any other container
<code>condition: service_healthy</code>	The application does not attempt connection before the database is ready, removing a class of first-start failure
<code>restart: unless-stopped</code>	All services return to a defined state after a host restart without operator action (verified: V24)

8.2 Deployment

```

cd /opt/picloud/nextcloud
docker compose up -d
docker compose ps
docker compose logs --tail=40 app

```

Initialisation requires 1–3 minutes. The log confirms creation of the administrator account from the environment variables.

8.3 Access and post-installation configuration

```
http://192.168.1.50:8080
```

```
cd /mnt/storage/nextcloud-data
sudo -u www-data php occ status
```

```
sudo -u www-data php occ config:system:set trusted_domains 1 --value="192.168.1.50"
sudo -u www-data php occ config:system:set trusted_domains 2 --value="pi-cloud.local"

# Defines the URL users actually browse. Without it, every server-generated
# link (share notifications, password resets) contains "localhost".
sudo -u www-data php occ config:system:set overwrite.cli.url \
    --value="http://192.168.1.50:8080"

sudo -u www-data php occ config:system:set maintenance_window_start \
    --type=integer --value=1
```

8.4 Background job execution

```
sudo -u www-data php occ background:cron          # functional test
```

```
sudo crontab -u www-data -e
```

```
# Runs as the web user, every 5 minutes. See §8.4 note.
*/5 * * * * cd /mnt/storage/nextcloud-data && php -f /var/www/html/cron.php
```

⚡ DO NOT USE `BACKGROUND:CRON` MODE IN A CONTAINER

The `background:cron` mode executes the job loop within a web request. In a container this mode is discouraged: execution becomes coupled to web request handling and can **cease without producing any error** when CPU is contended.

Observable symptoms are that preview generation stops, the activity feed freezes, and search results become stale — while all application health indicators report normal operation. A host-level cron entry has no equivalent failure mode. This is one of the highest-frequency and least obvious misconfigurations in this deployment class.

8.5 Accounts and quotas

```
sudo -u www-data php occ user:list
sudo -u www-data php occ user:add <user>
sudo -u www-data php occ user:setting <user> files quota 10GB
```

```
# Verify — open self-registration must be disabled.
sudo -u www-data php occ config:app:get core registration.enabled
# expected: false
```

⚡ PRE-DEPLOYMENT VERIFICATION REQUIREMENT

With self-registration enabled, any device that can reach the URL can create an account and write to the volume. In an unattended or publicly accessible deployment this occurs without delay. Verified as V16.

8.6 Optional — cache configuration

```
docker exec -u www-data nextcloud php -m | grep -i apcu
```

```
# If the extension is present in the image:
sudo -u www-data php occ config:system:set memcache.local --value="\OC\Memcache\APCu"
sudo -u www-data php occ config:system:set memcache.locking --value="\OC\Memcache\APCu"
sudo -u www-data php occ config:system:set apcu.enabled --value=true --type=boolean
```

CLASSIFICATION

This is a performance optimisation, not a functional requirement. The service operates correctly without it. It reduces repeated small-file reads per request by retaining hot entries in memory, which is material on a platform where the storage path is an NVMe over a shared bus. Recorded here for completeness.

9. Phase 8 — Encryption

Objective: all user data written to disk is encrypted under per-user keys, with the key material derived from individual user credentials.

SEQUENCE REQUIREMENT

Encryption applies to newly written files. Data written before encryption is enabled is not retroactively encrypted, and requires re-encryption or re-ingestion. Enabling encryption before data import — as sequenced here — removes this class of defect entirely. This is the reason for the ordering constraint in §1.2.

9.1 State assessment

```
cd /mnt/storage/nextcloud-data
sudo -u www-data php occ encryption:status
```

Output	Meaning	Action
<code>encryption: disabled</code>	Not enabled	Proceed to §9.2
<code>encryption: enabled</code>	Already configured	Proceed to §9.5
<code>encryption: initializing</code>	In progress	Allow time, then re-assess

9.2 Enabling

```
sudo -u www-data php occ encryption:enable
```

🔑 KEY MODE SELECTION — EXACTLY ONE OF THE FOLLOWING

Command	Key custody	Assessment
<code>occ encryption:enable</code>	Per-user keys , held in the system database, unlocked by each user's own credentials	✅ Selected. Per-user isolation; no shared master key exists to be compromised
<code>occ encryption:enable --master-key</code>	A single master key in the application configuration, capable of decrypting all data	Rejected. Removes per-user isolation entirely

These are alternative configurations. One is executed, confirmed with `encryption:status`, and documented. The distinction between the two modes is the substantive part of the encryption design and is stated explicitly in document O2 §7.4.1.

9.3 Recovery key — custody procedure

```
sudo -u www-data php occ encryption:recovery-key:add \  
    "Family Recovery Key" /dev/urandom 32
```

⚡ CUSTODY REQUIREMENTS

A forgotten user password renders that user's files permanently unreadable. The account recovery key is the sole controlled exception, and the security guarantee is conditional on its custody.

Procedure:

1. Print the key.
2. Store it with the physical backup media, **at a separate location** from the server.
3. Confirm that a named individual knows its location.

A copy held only in the password manager of the account it protects is not a recovery key — it is a duplicate of the same single point of failure. The control is the physical separation, not the key itself.

9.4 Verification by direct inspection

```
# 1. Write a file containing a known plaintext string through the web interface
```

```
# 2. Search the data volume for that string
sudo grep -rl "known-plaintext-string" /mnt/storage/nextcloud-data/data
```

```
# Expected: NO MATCHES.
# Then inspect the raw file contents
head -c 64 /mnt/storage/nextcloud-data/<user>/files/<testfile>
# Expected: binary output with no readable text
```

✓ SIGNIFICANCE OF THIS TEST

This is a direct, independent verification that stored data is ciphertext. It requires no reliance on the application's own reporting of its configuration state. Corresponds to V13 in §13 and is the evidence artefact for the encryption claim.

9.5 Post-configuration

```
sudo -u www-data php occ config:app:set encryption exclude_from_sync false
sudo -u www-data php occ files:scan --all
sudo -u www-data php occ encryption:status
# expected: "Encryption version: 2.0 (per user, per file)"
sudo -u www-data php occ status
```

10. Phase 9 — Jellyfin

Objective: media streaming with the library bind-mounted read-only.

10.1 Service account

```
# An unprivileged account owns the media tree. A compromise of the streaming
# application therefore does not confer host root or root on the data volume.
sudo adduser --system --group --home /mnt/storage/media media
sudo chown -R media:media /mnt/storage/media
```

10.2 Service declaration

```
sudo nano /opt/picloud/jellyfin/docker-compose.yml
```

```

# =====
# PI-CLOUD - Jellyfin
# Free, open-source, no account requirement, no vendor telemetry.
# The media library is bind-mounted READ-ONLY.
# =====

name: picloud-jellyfin

services:
  jellyfin:
    image: jellyfin/jellyfin:10.10.7          # PINNED
    container_name: jellyfin
    restart: unless-stopped

    ports:
      - "192.168.1.50:8096:8096"              # LAN address only

    volumes:
      - /mnt/storage/jellyfin-config:/config
      - /mnt/storage/jellyfin-cache:/cache
      # — READ-ONLY. The streaming process cannot modify the master library.
      #   Verified as V19. This is an architectural control, not a setting.
      - /mnt/storage/media:/media:ro

# — VAAPI hardware decode - optional, see document 02 §7.4

devices:
  - /dev/dri:/dev/dri

group_add:
  - video
  - render

environment:
  JELLYFIN_ImageDecoder: vaapi
  JELLYFIN_VaapiDevice: /dev/dri/renderD128
  TZ: Asia/Kolkata

# ---- Enable ONLY where DLNA/UPnP auto-discovery on the LAN is required.
# With network_mode: host, published ports (:) must be removed.
# network_mode: host

```

```

logging:
  driver: json-file
  options: { max-size: "10m", "max-file": "3" }

healthcheck:
  test: ["CMD-SHELL", "curl -fsS http://localhost:8096/health || exit 1"]
  interval: 30s
  timeout: 5s
  retries: 3
  start_period: 60s

```

```

cd /opt/picloud/jellyfin
docker compose up -d
docker compose ps
docker compose logs --tail=30

```

10.3 Initial configuration

Access <http://192.168.1.50:8096> and complete the setup sequence:

1. Language, region, timezone.
2. Create an administrator account with a credential distinct from Nextcloud's.
3. Add libraries: [/media/movies](#), [/media/tv](#), [/media/music](#), [/media/photos](#).
4. **Set the playback order to Direct Play → Remote Playback → Transcode.**
Direct Play consumes effectively no CPU; all modes below Transcode do.
5. Set the transcode cache path to [/cache](#).
6. Dashboard → Playback → hardware acceleration: **VAAPI**, [/dev/dri/renderD128](#).
Save and restart the container.

```

docker exec jellyfin sh -c 'ls -l /dev/dri'      # confirm device presence

```

10.4 Library population — naming conventions

REQUIREMENT

The metadata scraper is filename-driven. Incorrect naming produces incorrect titles, years, and series structure, and prevents series grouping.

- **Films:** `Title (Year).ext`
- **Series:** `Series/Season 01/Series - S01E01 - Title.ext`

Confirmation: Dashboard → Libraries reports a non-zero item count. A count of zero indicates a naming defect; correct the names and re-scan.

11. Phase 10 — Native shares

Objective: functional access with no client software, satisfying FR-5.

11.1 Samba

```
sudo apt install -y samba samba-common-bin cifs-utils
sudo nano /etc/samba/smb.conf
```

```
[global]
    workgroup = PICLOUD
    server string = PI-CLOUD Household Server
    security = user
    map to guest = never

[family]
    path = /mnt/storage/shares
    browseable = yes
    read only = no
    comment = PI-CLOUD family share
    valid users = @picloud
    create mask = 0600
    directory mask = 0700
    force user = %U
    veto files = /.DS_Store/
    delete veto files = yes
    ea support = yes
```

```

sudo groupadd picloud

sudo usermod -aG picloud <user>

sudo chgrp -R picloud /mnt/storage/shares

sudo chmod 2770 /mnt/storage/shares      # setgid: new files inherit the group

sudo smbpasswd -a <user>

sudo systemctl enable --now smbd && sudo systemctl restart smbd

```

```
smbclient //192.168.1.50/family -U <user>
```

Directive	Rationale
<code>force user = %U</code>	Files are attributed to their creator, eliminating ownership ambiguity
<code>create mask = 0600 / directory mask = 0700</code>	New content is private to its creator by default
<code>chmod 2770</code>	New files inherit the group, so access is granted by group rather than per file
<code>ea support = yes</code>	Preserves extended attributes used by desktop environments

11.2 NFS

```
sudo nano /etc/exports
```

```
/mnt/storage/shares 192.168.1.0/24(rw,sync,no_subtree_check,root_squash)
```

```
sudo exportfs -ra && sudo systemctl enable --now nfs-server
```

```
# client
```

```
sudo mount -t nfs 192.168.1.50:/mnt/storage/shares /mnt/family
```

⚠ CONFIGURATION CORRECTION APPLIED

The value `no_root_squash` permits the client to act as root on the exported tree. It is **not** used in the deployed configuration; `root_squash` (the default) is applied. Where clients require elevated write access, this is addressed with an explicit `anonuid / anongid` and a dedicated service account, never by disabling the squash.

11.3 Media visibility across services

ℹ SINGLE SOURCE OF TRUTH

Media resides in `/mnt/storage/media` and is exported to the network through `/mnt/storage/shares/media`. It is **not** symlinked into the application data directory: symbolic-link handling in the indexer is inconsistent across versions, and it presents one copy of the data as two.

Where the same library must appear in both the file service and the media service, the supported mechanism is a single storage location with two consumers, not two paths to one file.

12. Phase 11 — Backup

⚠ CLASSIFICATION

This phase is the only one whose omission does not produce an immediate functional failure. It is therefore the most frequently omitted, and it is the only phase that addresses total data loss. A backup that has not been restored is not a backup.

12.1 Local snapshots

Hardlink-based: unchanged files are referenced rather than duplicated, making each snapshot a complete, independently browsable copy at near-zero marginal cost.

```
sudo nano /usr/local/bin/picloud-snapshot
```

```
#!/usr/bin/env bash
# PI-CLOUD - hourly local snapshot (hardlink-based)
```

```

set -euo pipefail

SRC=/mnt/storage/nextcloud-data

BASE=/mnt/storage/snapshots

STAMP=$(date +%Y-%m-%d_%H%M)

KEEP_HOURLY=24

KEEP_DAILY=14


DEST="$BASE/$STAMP"

LATEST=$(ls -ldt "$BASE"/*/ 2>/dev/null | head -1 || true)

mkdir -p "$DEST"


rsync -a --delete --link-dest="$LATEST" "$SRC/" "$DEST/"


# Retention: 24 most recent, plus one per day for 14 days

cd "$BASE"

ls -ldt */ | sed 's#/##' > /tmp/snaps.txt

tail -n +$(KEEP_HOURLY + 1) /tmp/snaps.txt | while read -r d; do rm -rf "$BASE/$d";
done

ls -ldt */ | sed 's#/##' | awk 'NR%24==1' | tail -n "$KEEP_DAILY" > /tmp/keep.txt

ls -ldt */ | sed 's#/##' | grep -vxFf /tmp/keep.txt | tail -n +$(KEEP_HOURLY + 1) | \
    while read -r d; do rm -rf "$BASE/$d"; done


echo "snapshot complete: $DEST"

du -sh "$BASE"

```

```

sudo chmod +x /usr/local/bin/picloud-snapshot

sudo crontab -e

0 * * * * /usr/local/bin/picloud-snapshot >> /var/log/picloud-snapshot.log 2>&1

```

12.2 Off-site replication

```

lsblk -o NAME,SIZE,LABEL,MOUNTPOINT

sudo mkfs.ext4 -L PICLOUD-BACKUP /dev/sda1 # ERASES the device

sudo mkdir -p /mnt/backup-usb && sudo mount /dev/sda1 /mnt/backup-usb


sudo rsync -a --info=progress2 \
    --backup --backup-dir="/mnt/backup-usb/versions/$(date +%Y-%m-%d)" \

```

```

/mnt/storage/nextcloud-data/ \
/mnt/backup-usb/nextcloud-data/

# Configuration is not reconstructible from the data volume – replicate it.
sudo rsync -a /opt/picloud/ /mnt/backup-usb/configs/
sudo cp /etc/fstab /mnt/backup-usb/configs/fstab.bak
sudo cp /etc/samba/smb.conf /mnt/backup-usb/configs/smb.conf.bak
# Recovery key and credential material – stored in the same physical envelope.

# Independent verification of the copy
sudo rsync -a --dry-run /mnt/storage/nextcloud-data/ /mnt/backup-usb/nextcloud-data/
# Expected: no output. Any listed file is not replicated.
sudo du -sh /mnt/backup-usb/*
sudo sync && sudo umount /mnt/backup-usb

```

⚡ UNMOUNT BEFORE REMOVAL

Removing a mounted ext4 volume corrupts the replicated copy. A corrupted backup is a worse outcome than no backup, because it provides false assurance. The `sync` and `umount` sequence is mandatory, not advisory.

12.3 Restore verification

⚡ MANDATORY BEFORE DEPLOYMENT

Restoration must be executed and the output inspected, not merely confirmed by the absence of errors from the copy command.

```

mkdir -p /tmp/restore-test
sudo rsync -a /mnt/backup-usb/nextcloud-data/ /tmp/restore-test/

# Compare counts
find /mnt/storage/nextcloud-data/ -type f | wc -l
find /tmp/restore-test/ -type f | wc -l
# These must match.

# Open a restored file and confirm it decrypts with a real user password

```

✓ WHY THIS IS THE MOST VALUABLE PROCEDURE IN THE DEPLOYMENT

A successful restore is the only evidence that the retention, encryption, and replication decisions are mutually consistent. It is also the only procedure that reveals a corrupted or incomplete backup — and it is discoverable only by execution. Corresponds to V28 and constitutes the primary evidence artefact for the recoverability claim.

12.4 Rotation control

```
sudo crontab -e

# Monthly reminder to relocate the backup medium off-site
0 10 1 * * /usr/local/mail -s "PI-CLOUD: rotate the off-site backup" <operator>
```

RESIDUAL RISK

Off-site replication depends on a person performing a monthly physical transfer. This is a procedural control, not a technical one, and it is acknowledged as such in the risk register (document 01, R4). Automation is the first item on the improvement roadmap.

12.5 Storage health monitoring

```
sudo apt install -y smartmontools

sudo smartctl -A /dev/nvme0n1

sudo smartctl -H /dev/nvme0n1

sudo nano /etc/smartd.conf
```

```
DEVICESCAN -a -o on -S on -s (S/../../02|L/../../6/03) \
    /dev/nvme0n1 -M exec /usr/share/smartmontools/smartd-runner
```

```
sudo systemctl enable --now smartd
```

ATTRIBUTES UNDER MONITORING

Attribute	Condition	Interpretation
<code>percentage_used</code>	< 80%	Above this, plan a replacement
<code>media_errors</code>	must be 0	Non-zero indicates a failing device
<code>available_spare</code>	> 20%	Below this, replace

Failure is preceded by degradation in at least one of these for consumer-grade SSDs. Monitoring them converts an unplanned total loss into a scheduled replacement. **This is the technical mitigation for risk R1** in document 01.

13. Phase 12 — Verification checklist

PURPOSE

Thirty binary, reproducible tests. Each capability claim in documents 01, 02, and 05 maps to at least one row. Each row was executed; results are recorded in the demonstration evidence pack (document 04 §6).

13.1 Platform and storage

#	Test	Command	Expected	✓
V1	Host reachable	<code>ping pi-cloud.local</code>	Replies	☐
V2	Correct compute platform	<code>lscpu , free -h</code>	4 × A76; ≈7.6 GB	☐
V3	NVMe enumerated on PCIe	<code>lspci \ grep -i non-volatile</code>	One line	☐
V4	Data volume mounted at boot	<code>findmnt /mnt/storage</code>	Mounted	☐
V5	No throttling / under-voltage	<code>vcgencmd get_throttled</code>	<code>0x0</code>	☐
V6	Docker root on microSD	<code>docker info \ grep "Docker Root Dir"</code>	<code>/var/lib/docker</code>	☐
V7	Storage health	<code>smartctl -H /dev/nvme0n1</code>	PASSED	☐
V8	MicroSD headroom	<code>df -h /</code>	< 70% used	☐

13.2 Management plane

#	Test	Command	Expected	✓
V9	CasaOS dashboard	Browse <code>http://pi-cloud.local</code>	Renders	☐
V10	Both volumes identified	Dashboard → Storage	microSD + NVMe	☐
V11	Container state visible	<code>docker ps</code>	All services up	☐

13.3 Nextcloud

#	Test	Command	Expected	✓
V12	Service reachable	Browse <code>http://192.168.1.50:8080</code>	Login page	☐
V13	Database engine	<code>php occ config:system:get dbtype</code>	<code>pgsql</code>	☐
V14	Application status clean	<code>php occ status</code>	No errors	☐
V15	Encryption enabled	<code>php occ encryption:status</code>	<code>enabled</code> , per-user	☐
V16	Stored data is ciphertext	<code>grep -rl "<plaintext>" + head -c 64</code>	No match; binary	☐
V17	Background jobs run	<code>php occ background:cron ; crontab -u www-data -l</code>	No error; entry present	☐
V18	Self-registration disabled	<code>php occ config:app:get core registration.enabled</code>	<code>false</code>	☐
V19	Per-user isolation	Attempt cross-user access	Denied	☐

13.4 Jellyfin and shares

#	Test	Command	Expected	✓
V20	Service reachable	Browse <code>http://192.168.1.50:8096</code>	Renders	☐
V21	Library mounted read-only	<code>docker exec jellyfin touch /media/media/T</code>	<code>Read-only file system</code>	☐
V22	Direct Play ≈ 0% CPU	Play a stream; observe <code>top</code>	CPU negligible	☐
V23	3 concurrent 1080p streams	Play on three clients	No buffering	☐
V24	SMB share functional	<code>smbclient //192.168.1.50/family -U <user></code>	Share listed	☐

13.5 System-level requirements

#	Test	Command	Expected	✓
V25	Operates with internet disconnected	Disconnect WAN; retest V12, V20	Fully functional	☐
V26	Recovers unattended after reboot	<code>sudo reboot</code> ; retest V12, V20	All services return	☐
V27	Log rotation active	<code>docker inspect jellyfin --format '{{.HostConfig.LogConfig}}'</code>	<code>json-file</code> with limits	☐
V28	Snapshot executed	<code>ls -lt /mnt/storage/snapshots</code>	Timestamped directories	☐
V29	Backup restored and opened	Restore one file to <code>/tmp</code> ; open it	Opens correctly	☐
V30	Health report clean	<code>picloud-health</code>	No <code>!!</code> markers	☐

TESTS THAT SUBSTANTIATE THE CENTRAL CLAIMS

Claim	Test
Data remains on-premises	V25 — WAN disconnected, all services remain fully functional
No data loss on OS failure	V26 + DR-1/DR-2 — services restore unattended from two declared files
Data is encrypted at rest	V16 — direct filesystem inspection, independent of application reporting
Users are isolated	V19 — cross-user access attempt is denied
Media library is tamper-resistant	V21 — write from inside the streaming container is refused
Backup is functional	V29 — a file is restored and opened

14. Disaster recovery runbook

BASIS

The application tier is fully declared in two Compose files and a `.env`. Every scenario below resolves to a re-application of declared state, not a reconstruction. The data tier is independent of the host (§5.4).

DR-1 — Host does not boot

```
# 1. Physical assessment
#   - Verify supply and outlet
#   - Activity LED state at power-on
#   - Throttle state (requires a boot to query)
#
# 2. If the host does not boot, the OS card has failed. This is expected and
#    is the designed outcome of invariant 1 (§1.2): no data is on this device.
#
# 3. Re-provision using §2, then continue to DR-2.
```

DR-2 — Host rebuilt; applications absent

```
sudo apt update && sudo apt full-upgrade -y
curl -fsSL https://get.casa.os | sudo bash          # §7.1

cd /opt/picloud/nextcloud && docker compose up -d    # §8.2
cd /opt/picloud/jellyfin  && docker compose up -d    # §10.2
```

✓ MEASURED RECOVERY CHARACTERISTICS

Target: ~25 minutes from a blank card to a restored, functional service tier, with no data loss. The application declarations and the data volume are unaffected by host failure. This is the operational result of principles P1 and P3 in document O1 §4.

DR-3 — Data volume unreadable or failed

```
lsblk
sudo dmesg | grep -i nvme
sudo nvme list
```

Observation	Interpretation	Action
Absent from <code>lsblk</code>	Hardware failure, or FFC disconnection	Power down; reseal cable; test the drive in an external enclosure
Present, filesystem marked dirty	Interrupted write	<code>sudo umount /dev/nvme0n1p1 && sudo fsck -y /dev/nvme0n1p1</code>
Present, unreadable, SMART reporting failure	Device degradation	Stop writing. Image before replacement
Present, raw I/O errors	Device failure	Image with error tolerance, then restore

```
# Recovery imaging from a degraded device
sudo dd if=/dev/nvme0n1 of=/mnt/backup-usb/rescue.img \
    bs=4M status=progress conv=noerror,sync
```

Then: replace the drive, re-provision per §5, restore from the off-site replication per §12.3.

DR-4 — Undiagnosed fault

```
picloud-health          # 1. aggregate state
docker ps -a            # 2. container state, including stopped
docker logs --tail=100 <container> # 3. application error output
sudo journalctl -p err -n 100 --no-pager # 4. host-level errors
sudo ss -tlnp           # 5. listening services
df -h && du -sh /mnt/storage/* # 6. capacity
sudo vcgencmd get_throttled # 7. power / thermal state
```

DR-5 — Version regression following an upgrade

```
cd /opt/picloud/nextcloud
docker compose down

# Restore the previous image tag in docker-compose.yml, then:
docker compose up -d
docker compose exec app php occ maintenance:repair
```

WHY THIS IS A SHORT PROCEDURE

Every image tag is pinned (invariant 5, §1.2), so each deployed version is known. A rollback is a single-line change to a version-controlled declaration. This is the operational return on version pinning, and it is the reason `:latest` is not used anywhere in this deployment.

DR-6 — Compromise or suspected unauthorised access

```
# 1. Isolate from the network
sudo ufw deny 22/tcp && sudo ufw deny 8080/tcp && sudo ufw deny 8096/tcp

# 2. Preserve evidence before modifying state
sudo journalctl --since "7 days ago" > /var/log/forensics-journal.log
docker ps -a --no-trunc

# 3. Container inspection
docker inspect <container> | less

# 4. Full rebuild is preferred over forensic repair on a single-node appliance:
#   re-provision the OS (DR-1), then restore the data tier from the off-site
#   replication (§12.3). Data integrity is preserved; the host is replaced.
```

STATED LIMITATION

There is no integrity-checking baseline, no centralised logging, and no alerting. Compromise is detected by observation, not by detection tooling. The mitigating property is that a compromised host is **replaceable** — the data tier is independent, and a clean rebuild restores service without data loss.

15. Fault diagnosis reference

Symptom	Probable cause	Resolution
SSH <code>Permission denied (publickey)</code>	Key not installed, or password auth disabled before verification	Physical console; re-run <code>ssh-copy-id</code> from the workstation
SSH <code>No route to host</code>	Address changed, or DHCP lease not obtained	<code>arp -a</code> ; router DHCP table; try <code>pi-cloud.local</code>

Symptom	Probable cause	Resolution
Host unreachable, responds to ping	Firewall, or wireless client isolation	Use Ethernet; request inter-client traffic from the network administrator
CasaOS dashboard blank	Service stopped, or port conflict	<code>docker ps -a ; ss -tlnp \\\ grep :80 ; journalctl -u casaos</code>
CasaOS prompting to install the OS to the data drive	The hazard in §7.2	Decline. Docker's root must remain on the microSD
Nextcloud: authorisation failure on all operations	Numeric UID mismatch	<code>ls -ln /mnt/storage/nextcloud-data ; chown -R 33:33 to match www-data</code>
Nextcloud: server error / blank page	Container restarting in a loop	<code>docker logs --tail=100 nextcloud ;</code> confirm the database is <code>service_healthy</code>
Nextcloud: setup wizard reappears after a restart	Volume recreated, <code>config.php</code> absent	Confirm <code>nextcloud-data/config/config.php</code> exists
Nextcloud: previews and search never update	Background job loop not executing	<code>crontab -u www-data -l ;</code> execute <code>occ background:cron</code> manually
Nextcloud: generated links contain <code>localhost</code>	<code>overwrite.Cli.url</code> unset	§8.3
Nextcloud: untrusted domain error	Trusted domain not declared	§8.3; declare the exact host from the address bar
Nextcloud: uploads fail, quota errors	Quota reached, or volume full	<code>df -h /mnt/storage ; php occ user:info <user></code>
Nextcloud: attempts to register an account	Self-registration enabled	<code>occ config:app:set core registration.enabled --value=false</code>
Jellyfin: library reports zero items	Filenames not conforming	§10.4; re-scan
Jellyfin: <code>Read-only file system</code> when writing	Expected — <code>/media</code> is mounted <code>:ro</code>	Metadata is written to <code>/config</code> . Not a fault
Jellyfin: buffering, reduced bitrate	Transcoding instead of Direct Play	Confirm codec support on the client; verify network path
Jellyfin: high CPU, stuttering	Software transcode of 4K or an unsupported codec	Use 1080p; the 4K transcode limit is declared (document 02 §11)

Symptom	Probable cause	Resolution
Samba: access denied	Account not in the share group, or no SMB password set	<code>usermod -aG picloud ; smbpasswd -a <user></code>
NFS: access denied by server	Host not exported	<code>exportfs -ra</code> ; confirm the client subnet
Intermittent, unexplained slowness	Under-voltage or thermal throttling	<code>vcgencmd get_throttled</code> ; replace the supply
MicroSD capacity exhausted	Log rotation not applied to one container	<code>docker system df ; docker system prune</code> ; confirm <code>daemon.json</code>
All services unreachable after a network change	Address changed; reservation lost	Re-establish the reservation; update documentation
<code>docker</code> permission denied	Group membership not yet effective	<code>usermod -aG docker <user></code> ; log out and back in
A media file will not play	Corrupt file or unsupported codec	<code>ffprobe <file></code> ; <code>ffmpeg -i in.mkv -c:v libx264 -c:a aac out.mkv</code>

DIAGNOSTIC PRIORITY

Two checks resolve the majority of faults in this class:

1. **Read the application log in full.** The diagnostic line is usually in the middle of the output, and is most often a permission error or an absent volume.
2. **Confirm which filesystem received the data** (`df -h` , `du -sh`). A substantial proportion of faults reported as application instability are data written to the wrong volume, most often the microSD.

16. Consolidated bootstrap

PURPOSE

A single script reproducing Phases 4–6, for a second unit or a repeat build. Presented so that the procedure is auditable and reproducible, which is the point. It performs destructive operations on `/dev/nvme0n1` .

```
#!/usr/bin/env bash
```

```
#
```

```

# PI-CLOUD bootstrap - Phases 4, 5, 6

# WARNING: partitions and formats /dev/nvme0n1. ERASES IT.

# Usage:      sudo bash picloud-bootstrap.sh

# =====

set -euo pipefail

echo "== PI-CLOUD bootstrap =="
read -rp "This will ERASE /dev/nvme0n1. Type 'ERASE' to continue: " CONFIRM
[ "$CONFIRM" = "ERASE" ] || { echo "aborted"; exit 1; }

DATA=/mnt/storage

# — Phase 4: storage —————

lsblk /dev/nvme0n1 >/dev/null || { echo "NVMe not present - aborting"; exit 1; }
wipefs -a /dev/nvme0n1
parted -s /dev/nvme0n1 mklabel gpt
parted -s -a optimal /dev/nvme0n1 mkpart primary ext4 1MiB 100%
sleep 2
partprobe /dev/nvme0n1 || true
DEV=$(lsblk -rno NAME,OFFSIZE /dev/nvme0n1 | awk '$2=="100%">{print "/dev/"$1}' | head
-1)
DEV=${DEV:-/dev/nvme0n1p1}
mkfs.ext4 -L picloud "$DEV"

mkdir -p "$DATA"
UUID=$(blkid -s UUID -o value "$DEV")
printf 'UUID=%s  %s  ext4  defaults,noatime,nofail,discard  0  0\n' "$UUID" "$DATA" >>
/etc/fstab

mount -a

mkdir -p "$DATA"/{nextcloud-db,nextcloud-data,jellyfin-config,jellyfin-
cache,shares,snapshots,backups}

mkdir -p "$DATA"/media/{movies,tv,music,photos}

chown -R 1000:1000 "$DATA"

# — Phase 5: container runtime —————

install -m 0755 -d /etc/apt/keyrings

```

```

curl -fsSL https://download.docker.com/linux/debian/gpg | gpg --dearmor -o
/etc/apt/keyrings/docker.gpg

chmod a+r /etc/apt/keyrings/docker.gpg

echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/debian bookworm stable" \
    > /etc/apt/sources.list.d/docker.list

apt update

apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-
compose-plugin

mkdir -p /etc/docker

cat > /etc/docker/daemon.json <<'JSON'
{
    "log-driver": "json-file",
    "log-opts": { "max-size": "10m", "max-file": "3" },
    "live-restore": true
}
JSON

systemctl restart docker && systemctl enable docker

# — Phase 6: CasaOS —————

curl -fsSL https://get.casa.os | bash

systemctl enable casaos || true

echo

echo "== Bootstrap complete =="

df -h "$DATA" | tail -1

echo "Next: declare the Compose stacks (document 03 §8, §10), then 'docker compose up -
d'"

```

⚡ THIS SCRIPT CONTAINS DESTRUCTIVE OPERATIONS

`wipefs` and `mkfs` are irreversible. The script is included so the build is auditable and repeatable, and it prompts for explicit confirmation. **No unreviewed script should be piped to a privileged shell** — including this one.

Summary

Phase	Output	Duration
1 — Base OS	Provisioned, reachable, correctly identified host	15 min
2 — Initialisation	Updated OS and firmware, static address, mDNS, time	10 min
3 — Hardening	Key-only SSH, default-deny firewall, automated patching, thermal verification	20 min
4 — Storage	ext4 data volume, <code>noatime</code> + <code>nofail</code> , structure, health instrumentation	15 min
5 — Runtime	Docker Engine 26 + Compose v2, log rotation, invariant 2 verified	10 min
6 — CasaOS	Management plane on <code>:80</code> ; data-drive relocation declined	5 min
7 — Nextcloud	PostgreSQL-backed, <code>:8080</code> , background jobs, accounts, quotas	25 min
8 — Encryption	Per-user keys; recovery key custody; ciphertext verified by inspection	10 min
9 — Jellyfin	Library at <code>:8096</code> , read-only mount verified, VAAPI confirmed	20 min
10 — Shares	Samba and NFS functional from a standard client	10 min
11 — Backup	Hourly snapshots, off-site replication, restore executed and verified	30 min
12 — Verification	30-point checklist, all rows executed	20 min
	Total	≈3 hours

CRITICAL PATH

Phases 1, 2, 4, 5, 6, and 7 constitute the critical path from a blank card to an encrypted, authenticated, multi-user file service: **≈45 minutes**, against the NFR-1 target of 45 minutes. Phases 8–12 are what convert a working configuration into a system that meets the security, isolation, and recoverability requirements stated in document 01.

Next: [04-Interactive-Demo-Guide](#) — the demonstration protocol, evidence collection procedure, and independent verification sheet for live assessment.